Volumes by Known Cross Sections and Shells
Answer Key

AP Calculus AB/BC   Applications of Integration

Quick Answers

1. $\frac{64}{3}$
2. $\frac{81\pi}{16}$
3. $\frac{81\pi}{2}$
4. $\frac{256}{3}$
5. $\frac{128\pi}{5}$
6. $\frac{128\sqrt{3}}{15}$, 14.78
7. 0, 2, $\frac{8\pi}{3}$
8. 8π

Curriculum

Level: AP Calculus AB/BC   Applications of Integration
FUN-6 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–4 of 5 across 11 tagged checks.

1. Square cross sections over the region under $y = x$, $0 \le x \le 4$.

Step 1. At position x the base runs from the x-axis up to $y = x$, so the slice is a square of side $s = x$ and area $A(x) = x^2$.

Step 2. $V = \displaystyle\int_0^4 x^2\,dx = \left[\frac{x^3}{3}\right]_0^4 = \frac{64}{3}$.

Step 3. Sanity check: the solid sits inside the $4 \times 4 \times 4$ box, and $\frac{64}{3} \approx 21.3$ is well under 64 ✓

Common wrong answer: 8 — that is the area of the triangular base, not a volume.

$\boxed{\frac{64}{3} \text{ cubic units}}$

2. Semicircular cross sections over the region under $y = \sqrt{x}$, $0 \le x \le 9$.

Step 1. The base at position x has height $\sqrt{x}$, and that segment is the *diameter* of the semicircle, so the radius is $\frac{1}{2}\sqrt{x}$ and $A(x) = \frac{1}{2}\pi r^2 = \frac{\pi}{8}\left(\sqrt{x}\right)^2 = \frac{\pi x}{8}$.

Step 2. $V = \displaystyle\int_0^9 \frac{\pi x}{8}\,dx = \frac{\pi}{8}\left[\frac{x^2}{2}\right]_0^9 = \frac{\pi}{8} \cdot \frac{81}{2}$.

Step 3. That is $\dfrac{81\pi}{16} \approx 15.9$.

Common wrong answer: about 63.6, from taking $\sqrt{x}$ as the radius rather than the diameter — which quadruples the area of every slice.

$\boxed{\frac{81\pi}{16} \text{ cubic units}}$

3. Shells: $y = x^2$ on $[0,3]$ about the y-axis.
Step 1. A vertical strip at x has radius x from the axis of revolution and height x^2, so the shell contributes $2\pi x \cdot x^2\,dx$.
Step 2. $V = \displaystyle\int_0^3 2\pi x^3\,dx = 2\pi\left[\frac{x^4}{4}\right]_0^3 = 2\pi \cdot \frac{81}{4}$.
Step 3. $V = \dfrac{81\pi}{2} \approx 127.2$.
Common wrong answer: 20.25 — the 2π circumference factor was dropped.

$$\boxed{\frac{81\pi}{2} \text{ cubic units}}$$

4. Square cross sections over the half-disc under $y = \sqrt{16 - x^2}$.
Step 1. The base is the upper half of the circle of radius 4, so x runs from -4 to 4 and the slice at x is a square of side $s = \sqrt{16 - x^2}$.
Step 2. $A(x) = s^2 = 16 - x^2$ — squaring removes the radical, which is why square cross sections on a circular base are pleasant.
Step 3. $V = \displaystyle\int_{-4}^4 \left(16 - x^2\right)dx = \left[16x - \frac{x^3}{3}\right]_{-4}^4 = \frac{256}{3} \approx 85.3$.
Common wrong answer: $8\pi \approx 25.1$, from integrating the side length instead of its square.

$$\boxed{\frac{256}{3} \text{ cubic units}}$$

5. Shells: $y = \sqrt{x}$ on $[0,4]$ about the y-axis.
Step 1. Radius x, height $\sqrt{x}$, so the integrand is $2\pi x\sqrt{x} = 2\pi x^{3/2}$.
Step 2. $V = \displaystyle\int_0^4 2\pi x^{3/2}\,dx = 2\pi\left[\frac{2}{5}x^{5/2}\right]_0^4 = 2\pi \cdot \frac{2}{5} \cdot 32$.
Step 3. $V = \dfrac{128\pi}{5} \approx 80.4$.
Common wrong answer: 8π — discs about the x-axis answer a different question entirely.

$$\boxed{\frac{128\pi}{5} \text{ cubic units}}$$

6. Equilateral-triangle cross sections over the region under $y = 4 - x^2$.
Step 1. The parabola meets the x-axis at $x = \pm 2$, and the slice at x is an equilateral triangle of side $s = 4 - x^2$, so $A(x) = \frac{\sqrt{3}}{4}\left(4 - x^2\right)^2$.
Step 2. $\displaystyle\int_{-2}^2 \left(4 - x^2\right)^2 dx = \int_{-2}^2 \left(16 - 8x^2 + x^4\right)dx = \frac{512}{15}$.
Step 3. Multiply by the constant: $V = \frac{\sqrt{3}}{4} \cdot \frac{512}{15} = \frac{128\sqrt{3}}{15}$.
Common wrong answer: 34.13, the square-cross-section volume — the $\frac{\sqrt{3}}{4}$ factor was never applied.

$$\boxed{\frac{128\sqrt{3}}{15} \approx 14.78 \text{ cubic units}}$$

7. Shells between $y = 2x$ and $y = x^2$ about the y-axis.
Step 1. The curves meet where $x^2 = 2x$, that is $x^2 - 2x = 0$, so $x(x-2) = 0$.
$\boxed{x = 0 \text{ and } x = 2}$
Step 2. On $[0, 2]$ the line is above the parabola, so the shell height is the gap $2x - x^2$, and the radius is x: $V = \displaystyle\int_0^2 2\pi x\left(2x - x^2\right)dx$.
Step 3. $= 2\pi\left[\frac{2x^3}{3} - \frac{x^4}{4}\right]_0^2 = 2\pi\left(\frac{16}{3} - 4\right) = 2\pi \cdot \frac{4}{3}$, so $V = \dfrac{8\pi}{3} \approx 8.4$.
Common wrong answer: about 33.5, from using $2x$ alone as the shell height and forgetting that the parabola cuts the region off from below. $\boxed{\dfrac{8\pi}{3} \text{ cubic units}}$

8. Choosing the method: $y = x^2$, the y-axis and $y = 4$, about the y-axis.
Step 1. Revolving about the y-axis with the region described by y as a function of x is the shell case: shells integrate in x, so no rewriting of $y = x^2$ as $x = \sqrt{y}$ is needed, and the region's left boundary (the y-axis) is the axis of revolution, so there is no inner radius to subtract.
Step 2. A strip at x runs from the parabola up to $y = 4$, so the height is $4 - x^2$; the region ends where $x^2 = 4$, at $x = 2$. Hence $V = \displaystyle\int_0^2 2\pi x\left(4 - x^2\right)dx$.
Step 3. $= 2\pi\left[2x^2 - \frac{x^4}{4}\right]_0^2 = 2\pi\left(8 - 4\right) = 8\pi \approx 25.1$.
Manual review: the written justification is an open response. Accept any argument that shells integrate in x so the given function needs no inversion, or that washers in y would require $x = \sqrt{y}$ first — both are correct reasons, and a student who sets up the washer integral in y correctly and gets 8π has also answered the question. $\boxed{8\pi \text{ cubic units}}$